SI305 | Winter 2026
Final Project
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SI305 Final Report: Physician Density and Socioeconomic Disparities in Michigan ZIP Codes (1990–2021)

Summary

Physicians in Michigan tend to practice in wealthier areas, leaving low-income communities underserved. Michigan has had a worsening physician shortage, with the state's chief medical officer directly naming physician maldistribution as a core access problem. Highest-income ZIP codes had consistently higher physician density from 1990 to 2021, with the gap between highest and lowest incomes consistently widening. Median family income was the strongest predictor of physician density among all demographic predictors examined, making income level a reliable indicator for identifying underserved communities.

Problem Statement

Access to healthcare services in the United States is not distributed equally across communities. Research has consistently shown that physicians and other healthcare providers tend to practice in wealthier areas rather than communities that need them the most. A study by Davis et al. found that the number of primary care physicians was 31% higher in the highest income counties compared to the lowest income counties in the United States, which shows a clear relationship between socioeconomic status and healthcare provider availability.[1]

Low-income communities face serious barriers that worsen this gap significantly. Lazar and Davenport found three main barriers that keep low-income families from getting adequate healthcare: lack of education about available resources, complications with health insurance coverage, and personal distrust of healthcare providers.[2] These barriers reinforce each other, which makes it harder for low-income families to access care over time. Low-income individuals are more likely to rely on Medicaid for coverage, which pays physicians at significantly lower rates than private insurance, and gives physicians less financial incentive to practice in low-income areas.[3]

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[1] Matthew A. Davis et al., "Supply of Healthcare Providers in Relation to County Socioeconomic and Health Status," *Journal of General Internal Medicine* 33 (2018): 412–414. https://doi.org/10.1007/s11606-017-4287-4

[2] Malerie Lazar and Lisa Davenport, "Barriers to Health Care Access for Low Income Families: A Review of Literature," 2018. https://doi.org/10.1080/07370016.2018.1404832

[3] Stephen Zuckerman, Aimee F. Williams, and Karen E. Stockley, "Trends in Medicaid Physician Fees, 2003–2008," *Health Affairs* 28, no. 3 (2009). https://doi.org/10.1377/hlthaff.28.3.w510

The federal government recognizes this problem through its Health Professional Shortage Area (HPSA) designation system, administered by the Health Resources and Services Administration (HRSA). HPSAs identify geographic areas, specific populations, and facilities experiencing a shortage of primary, dental, or mental health care providers. Low-income populations can qualify for their own HPSA designation. This means that economic status alone can create a healthcare shortage even in areas that otherwise have enough providers.[4]

Michigan is no exception. The state's own chief medical officer has named physician maldistribution a core access problem, and low-income ZIP codes face that directly.[5] This unequal distribution has real effects on people's health, as residents in underserved areas are more likely to put off care, turn to emergency rooms for basic medical needs,[2] and face worse health outcomes for conditions that are treatable if caught in time. Physician shortages in low-income areas reflect a healthcare market where financial incentives push physicians toward wealthier areas and away from the communities that need care most. Low-income families, uninsured individuals, and communities of color are most likely to live in these underserved ZIP codes and face the hardest time finding and seeing a physician.[4]

Research Questions

1. How has physician density changed across Michigan ZIP codes between 1990 and 2021, and does it differ by income level?
2. Which demographic characteristics, specifically poverty rate, median family income, disadvantage index, proportion Hispanic, proportion non-Hispanic Black, are most strongly associated with low physician density across Michigan ZIP codes in 2021?

RQ1 Analysis: How has physician density changed across Michigan ZIP codes between 1990 and 2021, and does it differ by income level?

To examine how physician density changed across Michigan ZIP codes from 1990 to 2021, ZIP codes were grouped into four income quartiles using median family income. Both variables came from two NaNDA datasets: physician density from a healthcare services dataset, and median family income from a socioeconomic dataset.[6, 7] For each quartile, average physician density was

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[4] U.S. Health Resources and Services Administration, "What Is Shortage Designation?" June 2023. https://bhw.hrsa.gov/workforce-shortage-areas/shortage-designation

[5] Michigan Department of Community Health, "Michigan Faces Serious Shortage of Physicians," May 4, 2009. https://www.michigan.gov/mdhhs/inside-mdhhs/newsroom/2009/05/04/michigan-faces-serious-shortage-of-physicians

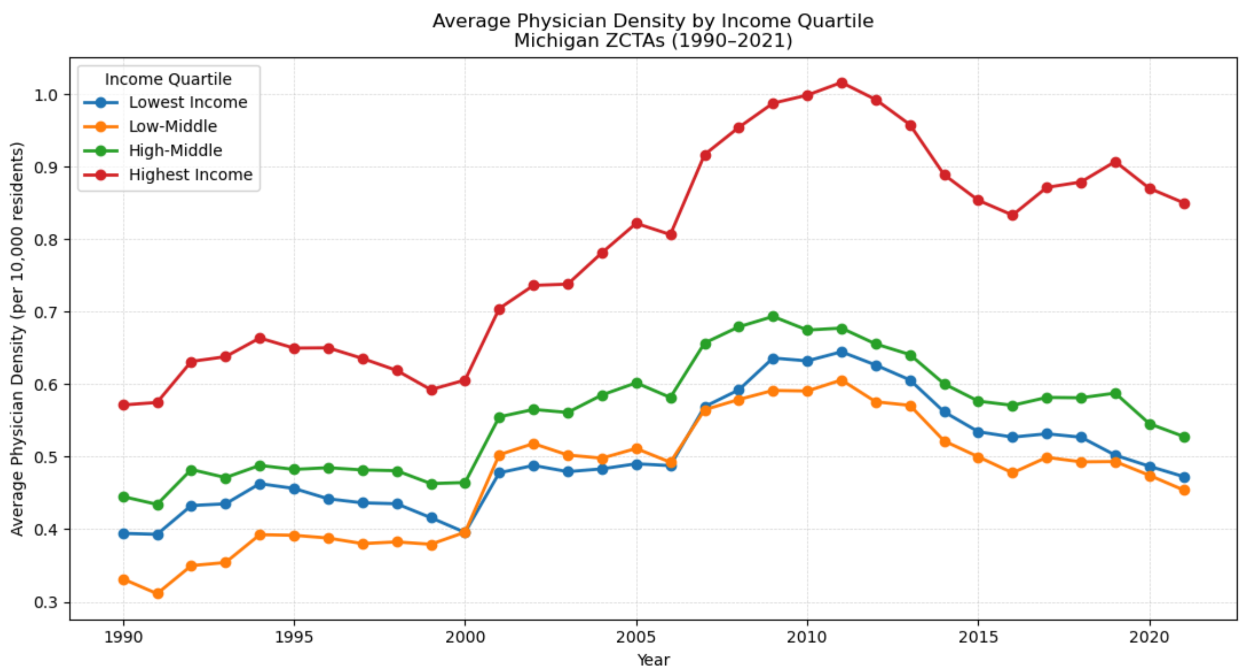
[6] University of Michigan Institute for Social Research, *NaNDA: Healthcare Services by Census Tract and ZCTA, United States, 1990–2022.* ICPSR.https://www.icpsr.umich.edu/sites/nanda/view/studies/209050

[7] University of Michigan Institute for Social Research, *NaNDA: Socioeconomic Status and Demographic Characteristics of ZCTAs, United States, 1990–2022.* ICPSR. https://doi.org/10.3886/ICPSR38528.v6

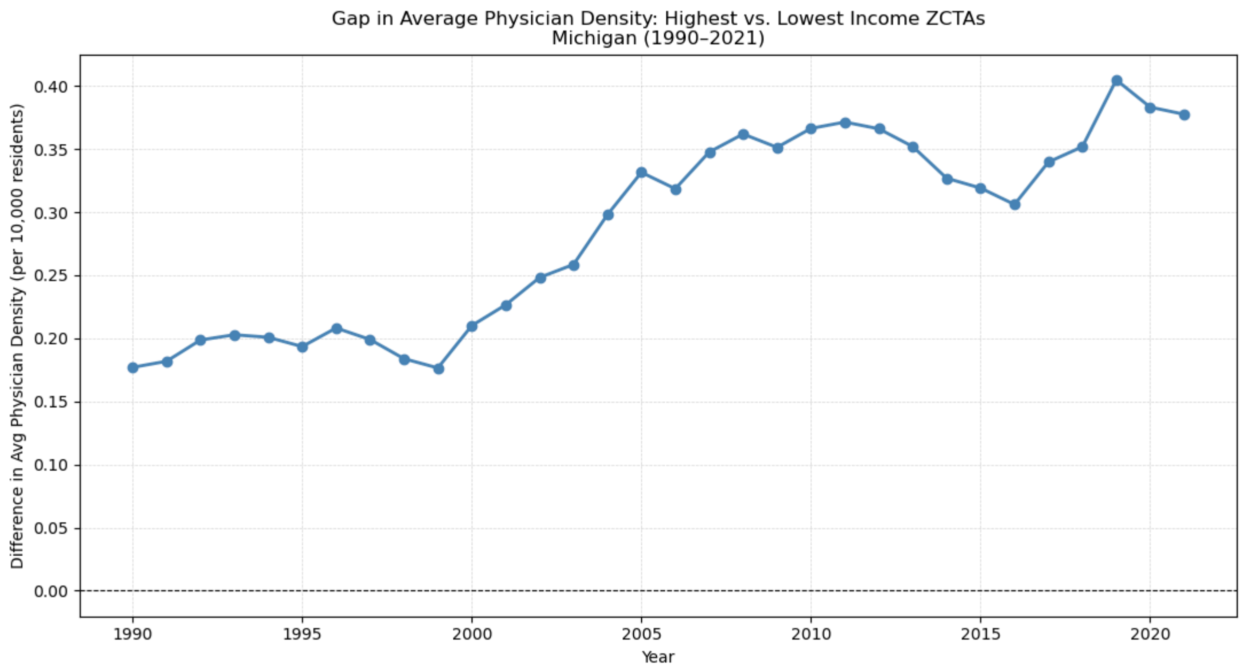
calculated for each year from 1990 to 2021 and compared across groups to identify whether trends differed by income level.

Highest income ZIP codes had consistently higher physician density than all other groups across the entire 31-year period. All four groups rose together from around 2000 to 2011, then declined after 2011. A closer look at the gap between the highest and lowest income quartiles showed that it never fully closed over the 31-year period, but doubled in size, narrowing briefly from 2013 to 2016, then widening again after 2016, peaking in 2019. One limitation of this analysis was that the income quartiles were based on a 2016–2020 snapshot, meaning ZIP codes that changed income level over the 31-year period may be misclassified. Additionally, physician density includes all specialties, not just primary care physicians, which may not fully reflect access to primary care for residents in underserved areas.

Visualizations:



Average physician density per 10,000 residents across Michigan ZIP codes from 1990–2021, grouped by income quartile.



Gap in average physician density between highest and lowest income quartiles across Michigan ZIP codes, 1990–2021.

Interpretation:

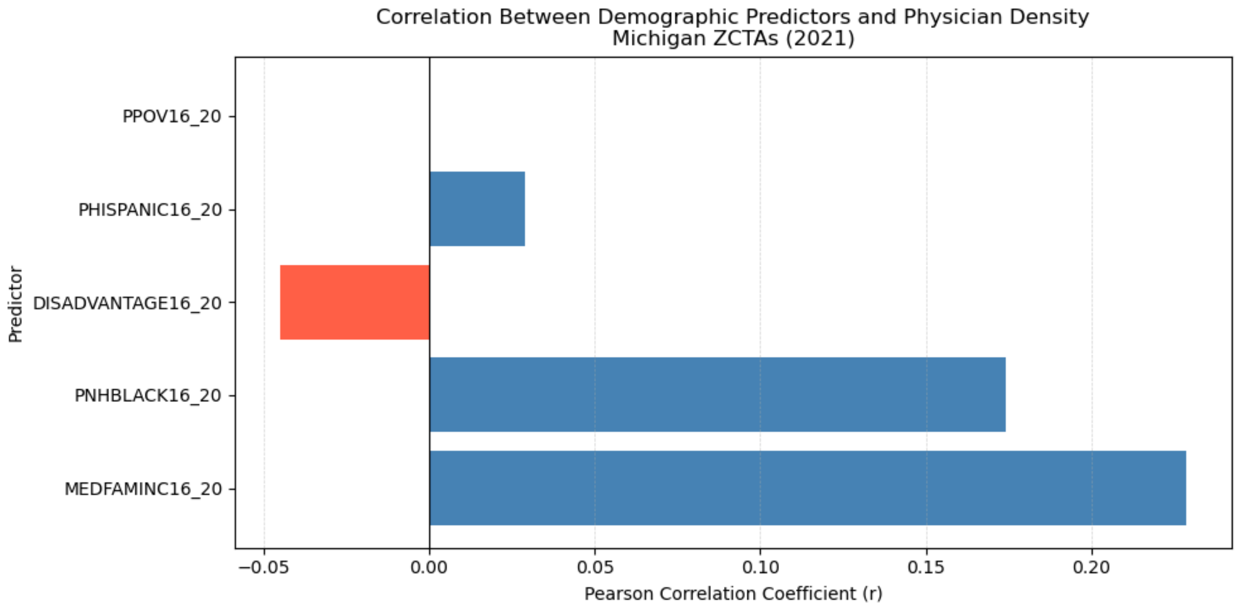
Across all four income quartiles, highest income ZIP codes had the greatest physician density throughout this entire 31-year period. This widening gap shows that low-income residents have had progressively fewer physicians serving their communities. This aligns with the pattern described in the problem statement, in which financial incentives consistently draw physicians toward wealthier areas. These findings motivated RQ2, which explores the specific demographic features associated with low physician density across Michigan ZIP codes in 2021.

RQ2 Analysis: Which demographic characteristics, specifically poverty rate, median family income, disadvantage index, proportion Hispanic, proportion non-Hispanic Black, are most strongly associated with low physician density across Michigan ZIP codes in 2021?

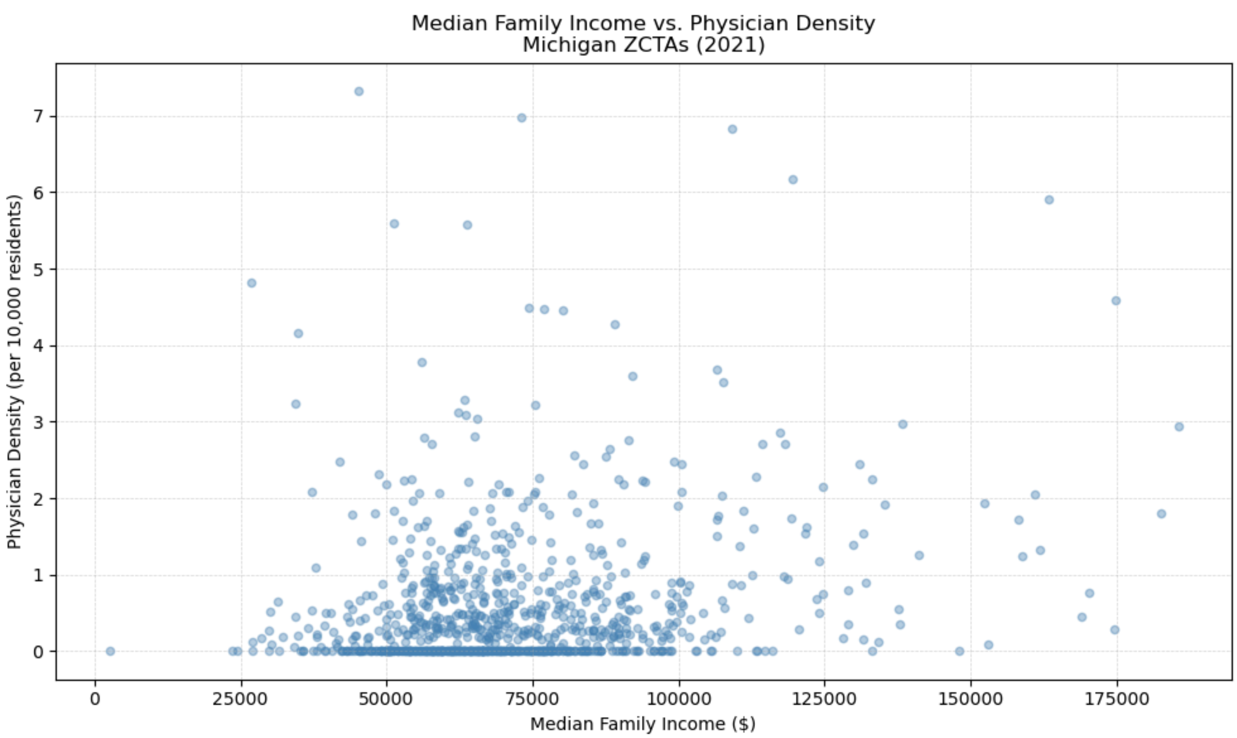
To identify which characteristics were most strongly associated with low physician density, the Pearson r correlation coefficient was calculated for predictors: poverty rate, median family income, disadvantage index, proportion Hispanic, proportion non-Hispanic Black. Physician density and socioeconomic variables were sourced from the same two NaNDA datasets used in RQ1.[6, 7]

The correlation analysis revealed that median family income had the strongest positive correlation ($r = 0.23$), and proportion non-Hispanic Black the second highest ($r = 0.17$). Both were statistically significant at $p < 0.001$. The remaining predictors had r values near zero, and showed weak associations with physician density. A limitation to consider was the multicollinearity between variables. Poverty rate had a near-zero correlation with physician density, likely because it overlaps so heavily with the disadvantage index, which had an r value of 0.92 between those two variables. The analysis is also limited to a single year, 2021, so it cannot capture how these relationships may have shifted over time.

Visualizations:



Correlation between demographic predictors and physician density per 10,000 residents across Michigan ZIP codes in 2021



Median family income vs. physician density per 10,000 residents across Michigan ZIP codes in 2021

Interpretation:

The association between median family income and physician density suggests that wealthier ZIP codes are better positioned to receive and keep physicians. Physicians have financial incentives to practice where patients have better insurance coverage and more ability to pay. The finding that areas with higher proportions of Black residents had more physicians on average was unexpected, but likely because Black residents in Michigan are concentrated in urban areas like Detroit, Flint, and Lansing, which also happen to have large hospital systems and other medical centers nearby. The scatter plot also shows considerable spread, which indicates that income alone does not determine physician access. Other factors, such as urban vs. rural location, and proximity to hospitals, also influence physician distribution.

Conclusion: Recommendations

Recommendation 1: Incentives for Physicians to Practice in Low-income ZIP codes

Audience: Michigan Department of Health and Human Services

RQ1 showed that over 31 years, the gap in physician density between high- and low-income ZIP codes has widened. RQ2 confirmed that median family income was the strongest predictor of physician density, reinforcing the idea that income is a clear indicator of low physician density. As the problem statement established, reduced payment rates give physicians less financial incentive to practice in low-income ZIP code areas. Initiating a medical school loan repayment program for physicians practicing in low-income ZIP codes would incentivize physicians to work in those underserved areas. While federal programs like HRSA's National Health Service Corps already exist, offering repayment of loans for physicians in underserved areas, a Michigan-specific program targeting the low-income ZIP codes would allow for more direct intervention. This would improve physician access in Michigan's most underserved communities.

Recommendation 2: Expand Telehealth Access in Low-income ZIP codes

Audience: Michigan Legislature

RQ1 and RQ2 both demonstrated that low-income ZIP code residents consistently have lower physician density. As stated in the problem statement, residents in underserved areas are more likely to put off care, turn to emergency rooms for basic medical needs, and face worse health outcomes for conditions that are treatable if caught in time. Since attracting physicians is a long term challenge, the Michigan Legislature should expand telehealth access in low-income ZIP codes, giving residents a way to receive preventive care without needing an in-person physician visit. The Legislature should allocate state funding to subsidize telehealth subscriptions or devices for these residents, making it easier to access routine and preventive care before conditions worsen.